Aim #12: How do we do transformations without the use of a coordinate plane?

Do Now:

Plot ΔABC with A(3,2), B(3,6), and C(6,2)

a) Reflect ΔABC over the x axis ($r_{x\text{-axis}}$)

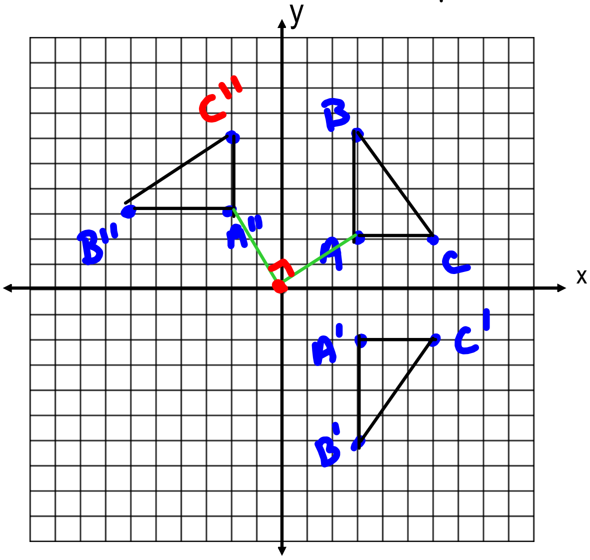
State the coordinates of the new triangle.

A' (3,-2) B'(3,-6) C'(6,-2)

b) Rotate ΔABC 90° counterclockwise around the origin ($R_{0,\,90}$)

State the coordinates of the new triangle.

A" (-2,3) B"(-6,3) C"(-2,6)

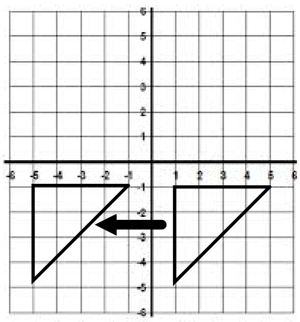


A transformation is a change in the location or size of a figure.

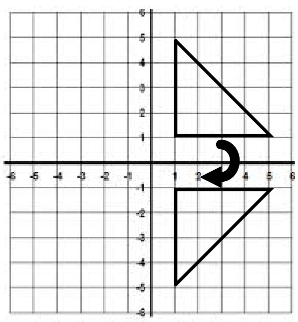
A **transformation**, F, of a plane is a **function** that assigns to each point P, the input, in the plane a unique point F(P), the output, in the plane.

We call a figure that is about to undergo a transformation the pre-image while the figure that has undergone the transformation is called the image.

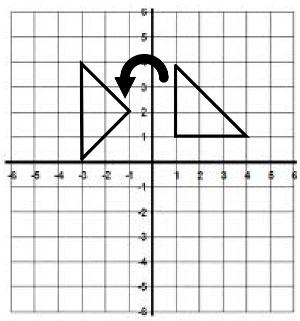
Name the type of transformation shown:



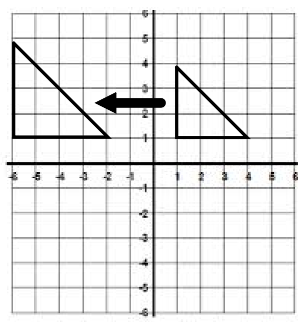
translation



reflection



rotation



dilation

Which of the above transformations keep side lengths and angle measures the same? First 3

Transformations that preserve **lengths of segments** and **measures of angles** are called basic rigid motions. They are: translations, rotations, reflections

Why is a dilation not a rigid motion? Use the facts above to clearly explain your answer. Length of segments is not preserved.

Rigid motions are also known as **isometric transformations** or **isometries**.
The pre-image and image following a rigid transformation are identical (congruent).
Link to read further about isometries:
https://en.wikipedia.org/wiki/Congruence_%28geometry%29#mediaviewer/File:Congruence_en.gif

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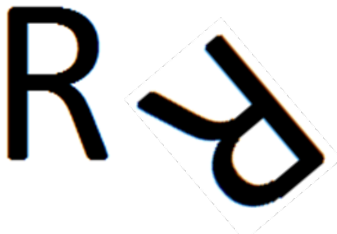
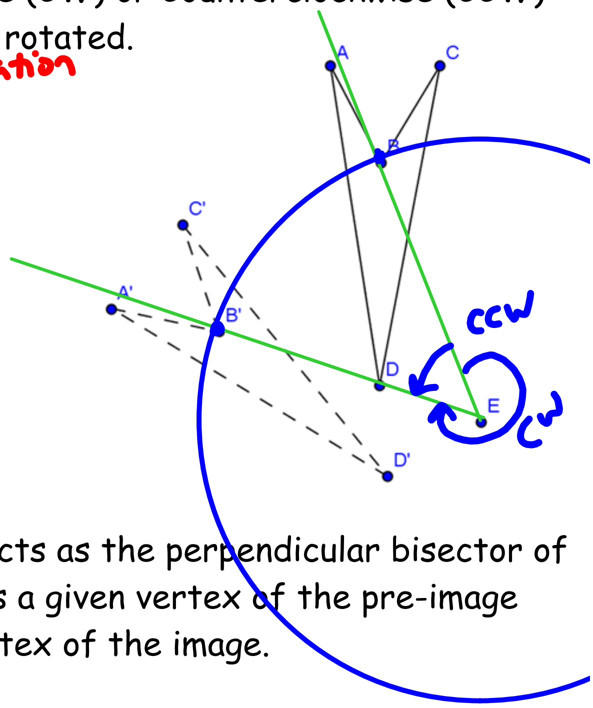
Rotation



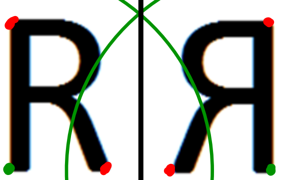
For a rotation, we need to know:
a) center of rotation ✓
b) direction: Clockwise (CW) or Counterclockwise (CCW)
c) number of degrees rotated. or the ∡ of rotation (name it)

Example: Use your protractor to find ∡BEB'
the angle of rotation and name it 3
ways: 50° (or 50° CCW), 310° CW, -310°
Note: Pre-image: (solid line)
Image: (dotted line)
Center of rotation: E



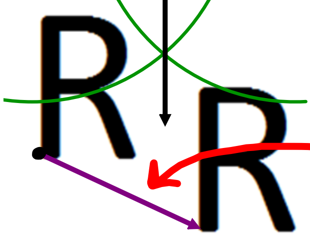
Reflection

← line of reflection



For a reflection:
The line of reflection acts as the perpendicular bisector of each segment that joins a given vertex of the pre-image with the respective vertex of the image.

Translation



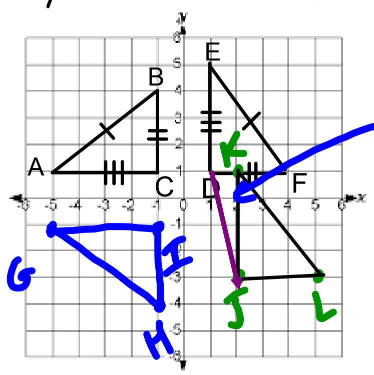
vector

For a translation, we need to know:
a) the point to be translated
b) the length and direction of the vector

270° or -90° or

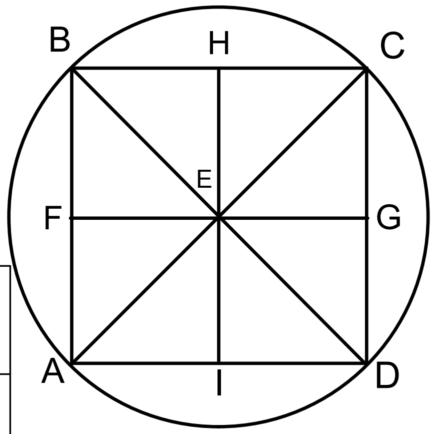
Example: a) By what rigid motion are ΔABC and ΔEFD congruent? rotation 90° CW
b) Draw and label ΔGHI congruent to ΔABC by reflection across the x-axis.
c) Draw and label ΔJKL congruent to ΔDEF by a translation 1 unit right and 4 units down, $T_{1,-4}$

rt/left, up/down



We will use the three rigid motions to find images.

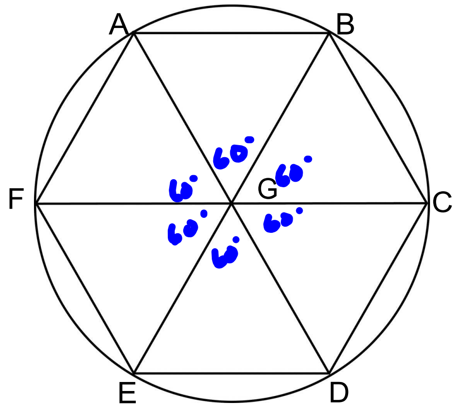
1. Square ABCD is inscribed in circle E (each vertex touches the circumference of circle E). Find the missing image.



same as 180° rot. about E

Pre-image	Transformation	Symbol	Image
point A	reflection over $\overleftrightarrow{FG}$	$r_{\overleftrightarrow{FG}}(A)$	B
point B	reflection over point E	$r_E(B)$	D
$\overline{BC}$	reflection over $\overline{FG}$	$r_{\overline{FG}}(\overline{BC})$	$\overline{AD}$ not $\overline{DA}$!
point D	using the same translation that maps C onto B	$T_?(D)$ $\overrightarrow{CB}$	A
$\overline{FG}$	translation along vector $\overrightarrow{EI}$	$T_{\overrightarrow{EI}}(\overline{FG})$	$\overline{AD}$
ΔHEC	reflection over $\overline{EG}$	$r_{\overline{EG}}(\Delta HEC)$	ΔIED
point H	rotation 90^0 clockwise about center E	$R_{E,-90^\circ}(H)$ $R_{E,90^\circ CW}(H)$	G
point D	rotation 90^0 counterclockwise about center E	$R_{E,90^\circ}(D)$	C
ΔABE	reflection over $\overline{HI}$	$r_{\overline{HI}}(\Delta ABE)$	ΔDCE
ΔABE	rotation 180^0 about center E	$R_{E,180^\circ}(\Delta ABE)$	ΔCDE
$\overline{AE}$	translation along vector $\overrightarrow{FH}$	$T_{\overrightarrow{FH}}(\overline{AE})$	$\overline{EC}$
point D	reflection over $\overleftrightarrow{AC}$	$r_{\overleftrightarrow{AC}}(D)$	B

2. ABCDEF is a regular hexagon (equal sides and equal angles), inscribed in a circle, center G.



$\frac{360}{6} = 60°$

a) What is the image of point A after a reflection over $\overleftrightarrow{FC}$?

$r_{\overleftrightarrow{FC}}(A) =$ E

b) What is the image of the point B after a reflection over point G?

$r_G(B) =$ E

c) What is the image of $\overline{BC}$ after a reflection over $\overleftrightarrow{AD}$?

$r_{\overleftrightarrow{AD}}(\overline{BC}) =$ FE

d) What is the image of point F after a reflection over the $\overleftrightarrow{BE}$?

$r_{\overleftrightarrow{BE}}(F) =$ D

e) What is the image of point B after a 60° counterclockwise rotation around G?

$R_{G,60°}(B) =$ A

f) What is the image of $\overline{CD}$ after a 180° counterclockwise rotation around G?

$R_{G,180°}(\overline{CD}) =$ FA

g) If the image of point F after a translation is A, what is the image of point E under
the same translation? G

h) What is the image of triangle GED after a reflection over $\overline{FC}$?

$r_{\overline{FC}}(\Delta GED) =$ ΔGAB

Geometry Assumptions

We have now done some work with all three rigid motions (rotations, reflections, and translations).

a. Any basic rigid motion preserves lines, rays, and segments. That is, for a basic rigid motion of the plane, the image of a line is a line, the image of a ray is a ray, and the image of a segment is a segment.

b. Any basic rigid motion preserves lengths of segments and angle measures of angles.

Let's Sum it Up!!

Activity: Use the three rigid transformations (isometries) in this activity to show shapes are congruent. http://www.learner.org/courses/teachingmath/grades3_5/session_02/section_02_b.html

Basic Rigid Motion: A basic rigid motion is a rotation, reflection, or translation of the plane. Given a transformation, the image of a point *A* is the point to which *A* is mapped by the transformation.

Distance Preserving: A transformation is said to be distance preserving if the distance between the images of two points is always equal to the distance between the pre-images of the two points.

Angle Preserving: A transformation is said to be angle preserving if

(1) the image of any angle is again an angle and

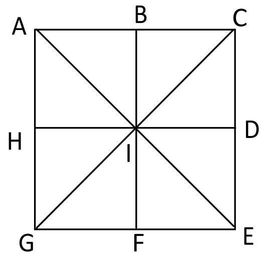
(2) for any given angle, the angle measure of the image of that angle is equal to the angle measure of the pre-image of that angle.

Name ________________________

Date _______________

CC Geometry H
HW #12

Answer the following questions based on the diagram below. ACEG is a square.



1) What is the image of point A after a reflection over $\overleftrightarrow{HD}$?
$r_{\overleftrightarrow{HD}}(A) =$_____

2) What is the image of point G after a reflection over the $\overleftrightarrow{BF}$?
$r_{\overleftrightarrow{BF}}(G) =$ ______

3) What is the image of the point C after a reflection over point I?
$r_I(C) =$ ______

4) What is the image of line segment $\overline{AG}$ after a reflection over the line $\overleftrightarrow{BF}$?
$r_{\overleftrightarrow{BF}}(\overline{AG}) =$ ______

5) What is the image of point A after a reflection over the $\overleftrightarrow{GC}$?
$r_{\overleftrightarrow{GC}}(A) =$ _______

6) What is the image of $\overline{GE}$ after a reflection over the line $\overleftrightarrow{HD}$?
$r_{\overleftrightarrow{HD}}(\overline{GE}) =$ ______

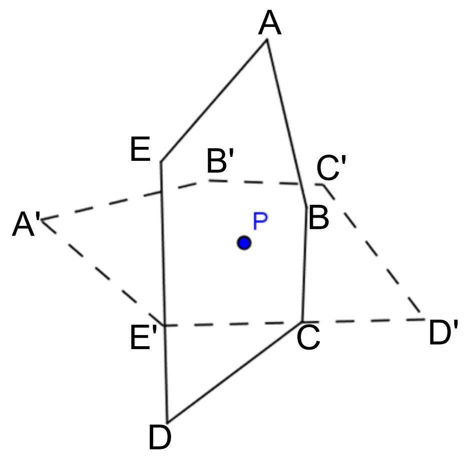
7) What is the image of point B after a 90° counterclockwise rotation around I? $R_{I,90^\circ}(B) =$ ______

8) What is the image of $\overline{CD}$ after a 180° counterclockwise rotation around I?
$R_{I,180^\circ}(\overline{CD}) =$_______

9) If the image of point A after a translation is I, what is the image of point H under the same translation? _______

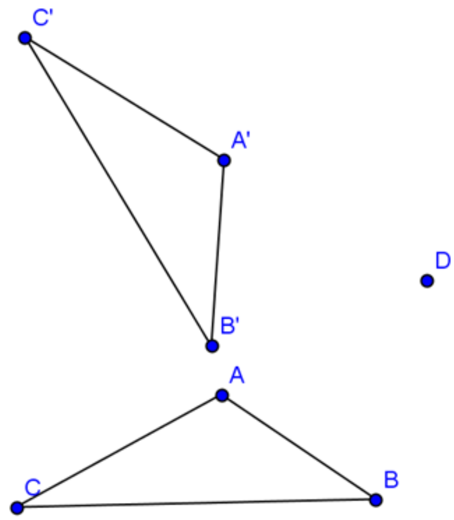
10) What is the image of ΔAHI after a reflection over $\overline{GC}$?
$r_{\overline{GC}}(\Delta AHI) =$______

11) Pre-image: (solid line)
Image: (dotted line)
Center of rotation: P



Name the angle of rotation three ways:________________________________

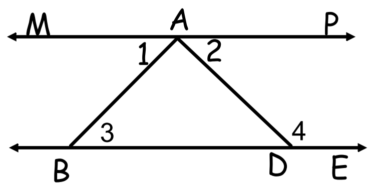
12) Pre-image: ΔABC
Image: $\Delta A'B'C'$
Center: D



Name the angle of rotation three ways:

____

Review:
Complete the following proof:



Given: $\overline{MAP} \parallel \overline{BDE}$, $m\angle 1 = m\angle 2$
Prove: $m\angle 3$ is supplementary to $m\angle 4$

statements	reasons
1. $\overline{MAP} \parallel \overline{BDE}$, $m\angle 1 = m\angle 2$	1. Givens